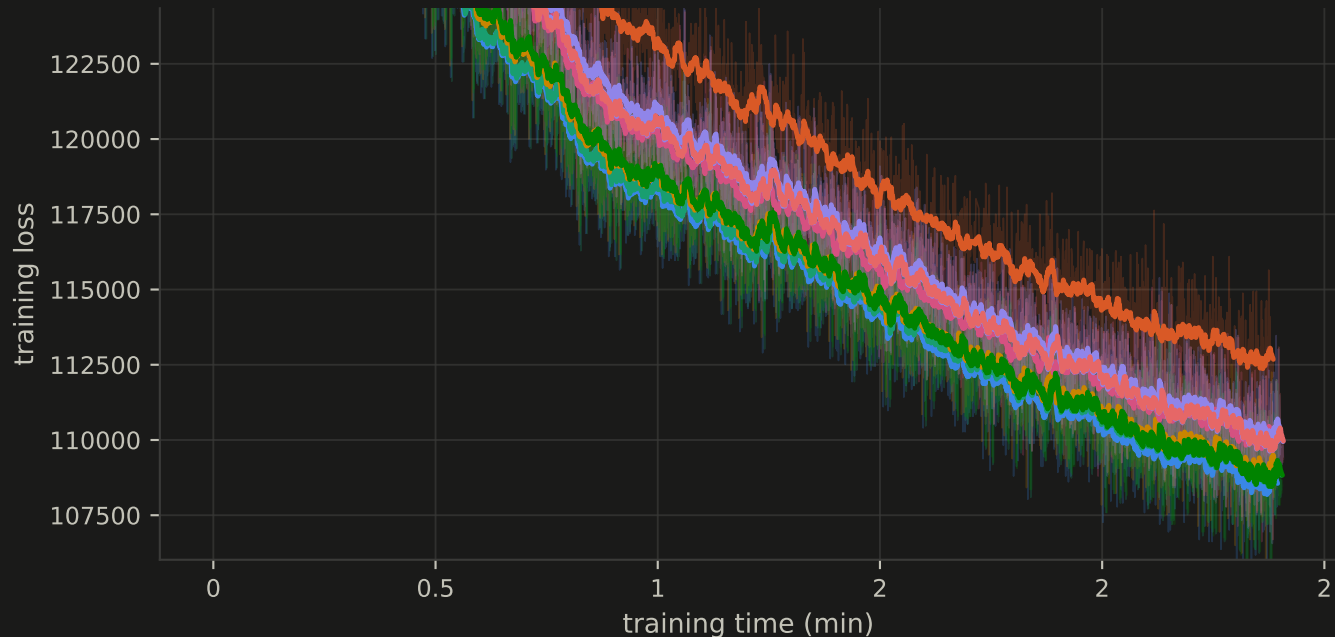


NanoGPT speedrun (record #40): training loss vs wall-clock



- | | | | |
|---------------------------|--------------------------------|---------------------------------|---------------------------------|
| Muon $\text{lr}=0.06$ | MuonUSign $\text{lr}=0.06$ | EF21-SignMuon $\text{lr}=0.06$ | EF21-MuonUSign $\text{lr}=0.06$ |
| SignSGD $\text{lr}=0.03$ | MuonSign $\text{lr}=0.03$ | EF21-MuonUSign $\text{lr}=0.06$ | EF21-MuonSign $\text{lr}=0.06$ |
| SignMuon $\text{lr}=0.03$ | EF21-SignMuon $\text{lr}=0.06$ | | |